Aakifah Minhaj

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Profile Summary

Computer Science undergraduate, specialising in AI/ML and full-stack development. Experienced in building and deploying AI-enabled applications with Python and LangGraph , alongside knowledge of GANs, VAEs, and Diffusion models. Strong foundation in Machine Learning, Cloud and Data analytics, with hands-on expertise in Generative AI and RAG systems

Education

Bits Pilani, Dubai Campus **Aug 2023 – July 2027**
Bachelor of Engineering in Computer Science *Dubai*

Experience

SmartLife **September 2025 – February 2026**
ERP Systems Developer Intern

- Cleaned and validated ERP inventory data (deduplication, null and format checks, unit normalization), reducing inconsistent records by 30%.
- Built a real-time Node.js/Express chatbot backend over WebSockets with an HTML/CSS/JS frontend, cutting latency by 40% compared with HTTP polling.
- Designed a JSON-driven finite-state machine for conversation states and transitions, so new flows need no code changes, reducing update effort by 50%.

Codelattice **June 2025 – August 2025**
Software Engineer Intern

- Automated invoice-to-barcode sticker generation by integrating Zoho Creator with Zoho Books APIs through Deluge, cutting label preparation time by 70%.
- Auto-extracted line-item fields (MPN, Size, MRP, Qty) into print-ready HTML/CSS templates sized to client labels, reducing labeling errors by 90%.
- Built Zoho Creator forms and workflows, and debugged API parsing, field mapping and barcode rendering, raising first-pass print success to 95%.

Projects

GenAI Solution Lab: Generative Model Training & Evaluation | *Python, PyTorch, Streamlit, PyTest*

- Built a reproducible PyTorch platform with checkpointing and CUDA/CPU support to benchmark GANs and VAEs on MNIST, using a pretrained classifier to evaluate sample quality.
- Implemented adversarial, reconstruction and KL losses, experiment tracking, a Streamlit dashboard, and a PyTest suite.

Enterprise Knowledge Assistant: RAG & LLM Architecture | *Python, FAISS, Ollama, SQLite, Streamlit*

- Built a RAG assistant over PDF/TXT/Markdown files with FAISS indexing that returns cited answers from a local Ollama LLM, with prompt-injection safeguards.
- Implemented hybrid (dense + BM25) retrieval, improving Recall@5 by 15% over dense-only search, evaluated with Precision@K, MRR, citation accuracy and latency.

scCoRGI: Deep Learning for Single-Cell Clustering | *PyTorch, TensorFlow, R, Scanpy*

- Designed a novel ZINB autoencoder with per-cell confidence weighting and GCN graph refinement before Leiden clustering.
- Benchmarked 11 methods (VAEs, autoencoders, graph models, NMF) on 38 datasets (500+ runs); scCoRGI reached 56.5% mean ARI and 83.6% purity, ranking 3rd and 2nd of 11.
- Beat VAE baselines scPhere by +10.6 ARI points (+23%) and DREAM by +15.9 points (+39%), confirmed by Friedman and Wilcoxon tests; fixed NaN and SVD failures in published code.

Publications

Next-Generation Wearable, Ingestible and Implantable Bioelectronics for Health Monitoring, Therapy and Human–Machine Interaction **Mar 2026**

Co-authored a peer-reviewed review in *Biomedical Microdevices*, Springer Nature DOI:10.1007/s10544-026-00809-x

Technical Skills

Languages: Python | JavaScript | Java | C | C++ | Assembly Language (MIPS)
Web Development: HTML | CSS (Tailwind & Bootstrap) | React.js
AI/ML: PyTorch | TensorFlow | scikit-learn | Deep Learning | NLP | GANs | VAEs
Generative AI & LLMs: Prompt Engineering | RAG | LLM Evaluation | LangGraph | Multimodal AI | Diffusion Models
Data Science & Analytics: Pandas | NumPy | Matplotlib | Seaborn | Power BI | Excel
Databases: MySQL | Microsoft SQL Server | MongoDB | Supabase
Cloud & DevOps: AWS | Azure | Docker
Tools & Platforms: Git/GitHub | Jupyter Notebook | VS Code